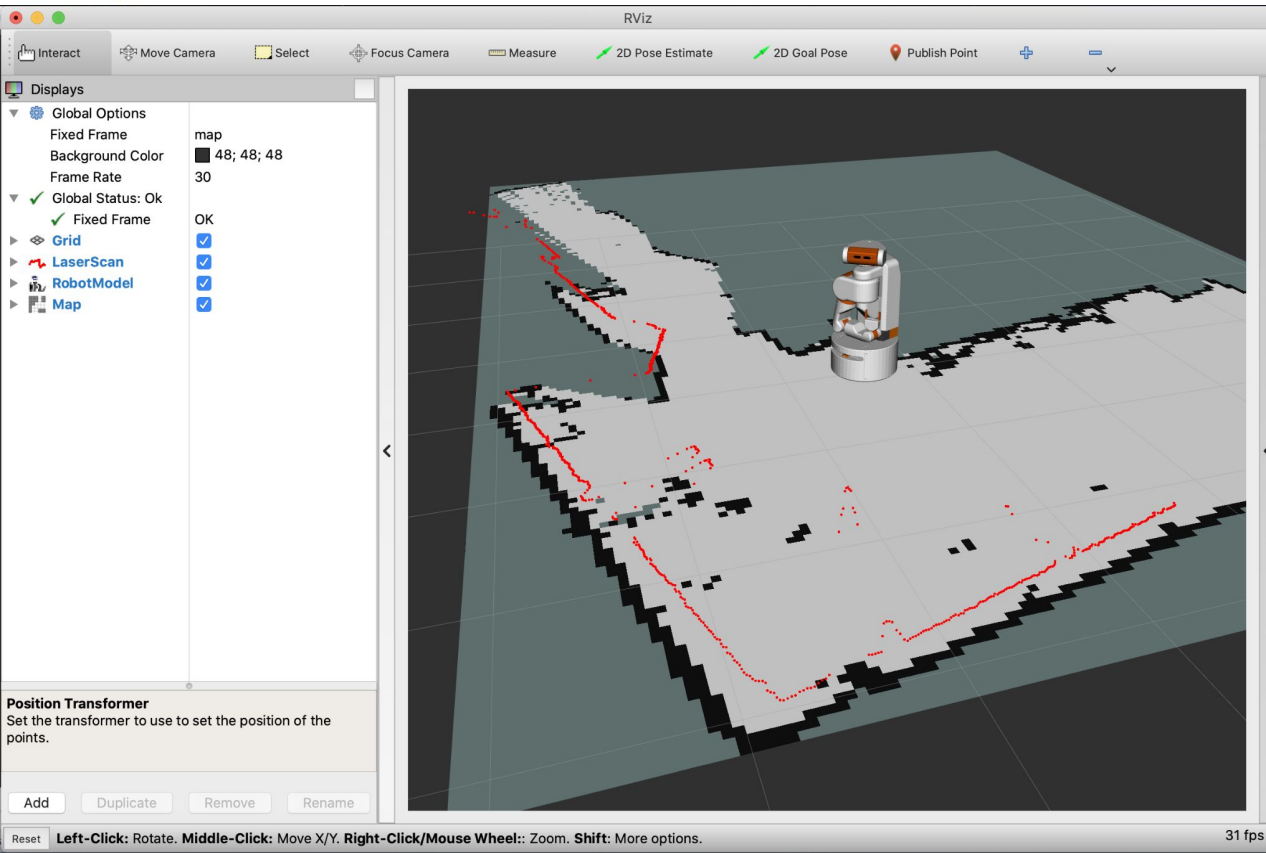


Mobile Robotics

Bachelor's
Degree in
Robotic
Intelligence

Maps



Maps

Maps are stored in a pair of files.

- The YAML file describes the map meta-data, and names the image file.

- The image file encodes the occupancy data.

The image describes the occupancy state of each cell of the world in the color of the corresponding pixel. In the standard configuration, whiter pixels are free, blacker pixels are occupied, and pixels in between are unknown.

YAML format

Required fields:

- **image** : Path to the image file containing the occupancy data; can be absolute, or relative to the location of the YAML file
- **resolution** : Resolution of the map, meters / pixel
- **origin** : The 2-D pose of the lower-left pixel in the map, as (x, y, yaw), with yaw as counterclockwise rotation (yaw=0 means no rotation).
- **occupied_thresh** : Pixels with occupancy probability greater than this threshold are considered completely occupied.
- **free_thresh** : Pixels with occupancy probability less than this threshold are considered completely free.
- **negate** : Whether the white/black free/occupied semantics should be reversed (interpretation of thresholds is unaffected)

Optional parameter:

- **mode** : Can have one of three values: trinary, scale, or raw. Trinary is the default.

Value Interpretation (1)

Given a pixel that has a value x in the range $[0, 256)$, how should we interpret this value when put into the ROS message? First we convert integer x to a floating point number p depending on the interpretation of the `negate` flag from the `yaml`.

- If `negate` is false, $p = (255 - x) / 255.0$.
This means that black (0) now has the highest value (1.0) and white (255) has the lowest (0.0).
- If `negate` is true, $p = x / 255.0$. This is the non-standard interpretation of images, which is why it is called `negate`, even though the math indicates that x is not negated (nomenclature is hard).

Value Interpretation (2)

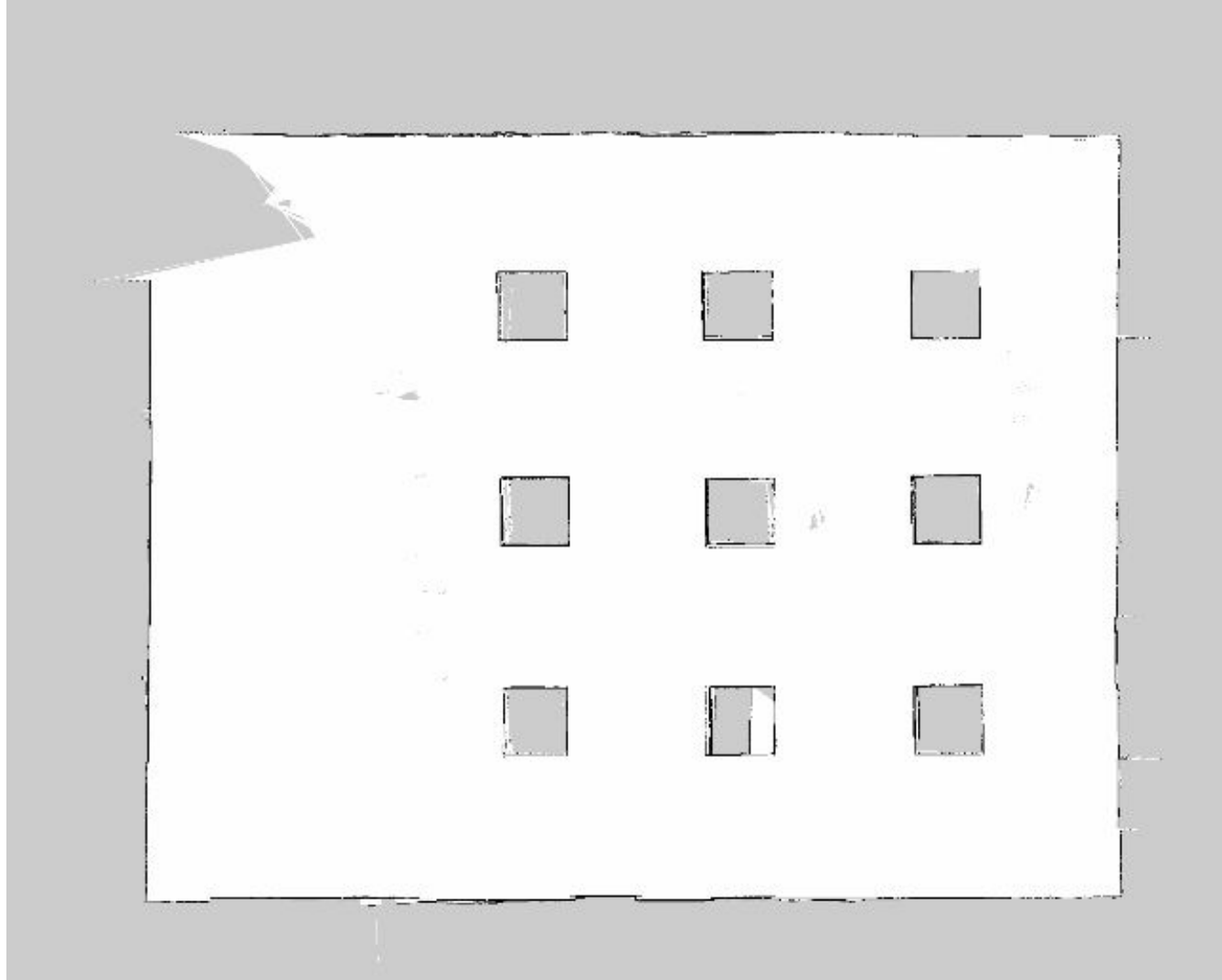
The standard interpretation is the trinary interpretation,
i.e. interpret all values so that the output ends up being one of three values:

- If $p > \text{occupied_thresh}$, output the value 100 to indicate the cell is **occupied**.
- If $p < \text{free_thresh}$, output the value 0 to indicate the cell is **free**.
- Otherwise, output -1 a.k.a. 255 (as an unsigned char), to indicate that the cell is **unknown**.

 occupied

 free

 unknown



Value Interpretation (3)

Scale mode

This tweaks the above interpretation to allow for more output values than trinary.

- As before if $p > \text{occupied_thresh}$, output the value 100.
If $p < \text{free_thresh}$, output the value 0.
- Otherwise, output $99 * (p - \text{free_thresh}) / (\text{occupied_thresh} - \text{free_thresh})$

This will allow you to output a full gradient of values ranging from $[0, 100]$. To output -1, simply use the alpha channel of a png, where any transparency will be interpreted as unknown.

Raw mode

This mode will output x for each pixel, so output values are $[0, 255]$.

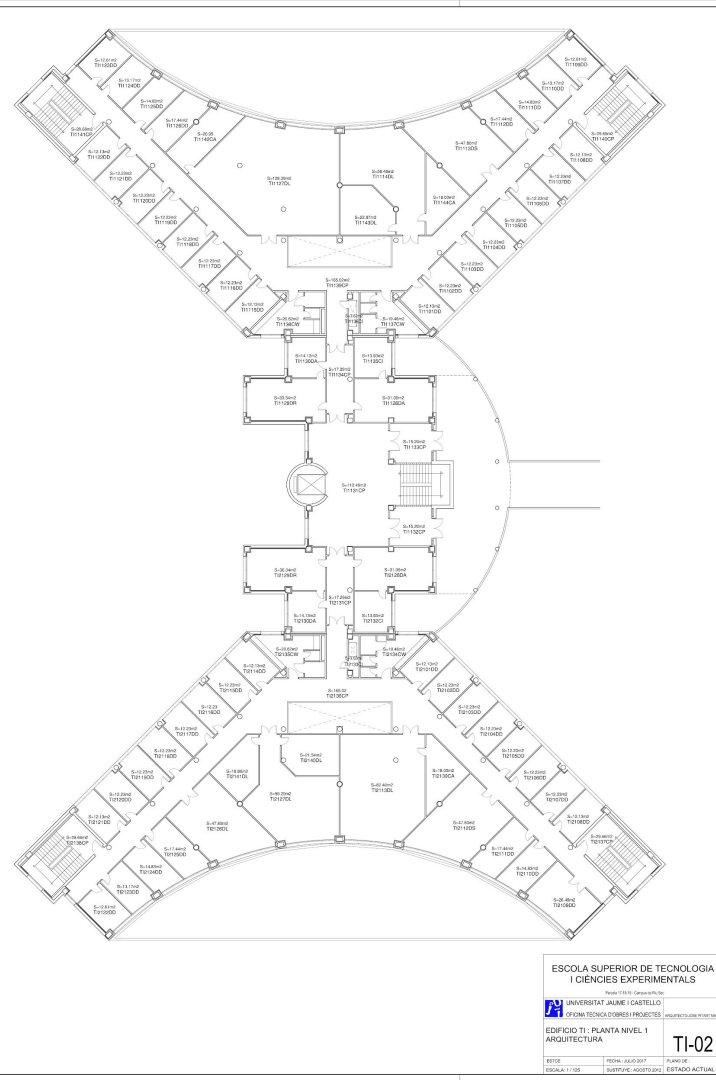
Task

Convert architectural map to ROS map:

1. Convert PDF to image
2. Adjust image size (optional)
3. Compute resolution
4. Edit image (delete names, doors, etc.)

TI_n1.pdf

Edificis de l'Escola Superior de Tecnologia i Ciències Experimentals





ESTADO ACTUAL

Tl_n1.pdf

Format: PDF-1.7

Paper size: A1, vertical (594 × 841 mm)

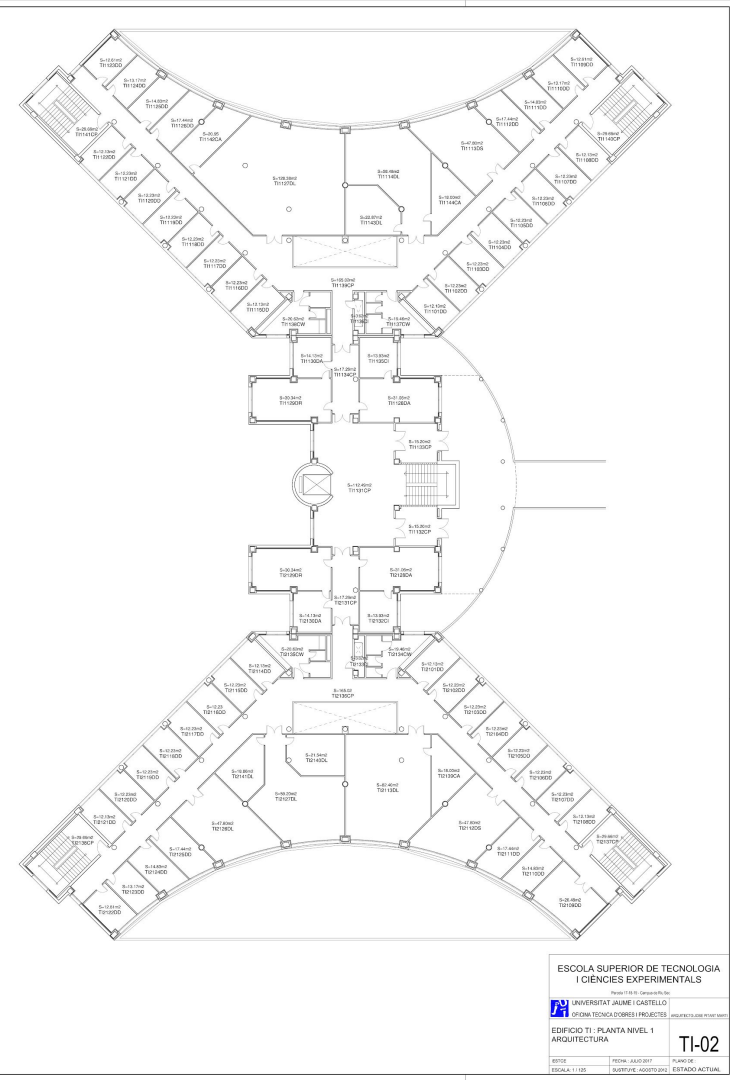
Scale: 1/125

PDF to PNG

Format: png (PNG)

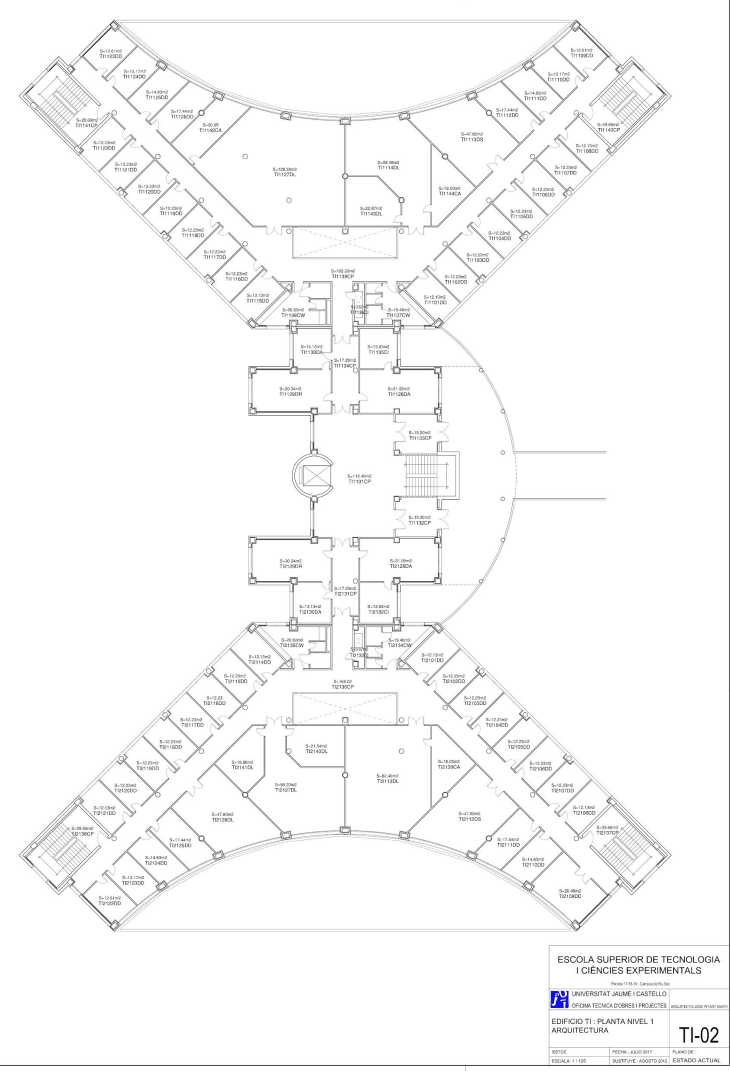
Width: 4678 px

Height: 6623 px



TI_n1.yaml

```
image: ./TI_n1.png
mode: trinary
resolution: meters / pixel ???
origin: [ , , 0]
negate: 0
occupied_thresh: 0.65
free_thresh: 0.25
```



Tl_n1.yaml

```
image: ./TI_n1.png
```

mode: trinary

resolution:

horizontal: $0.594 * 125 / 4678$

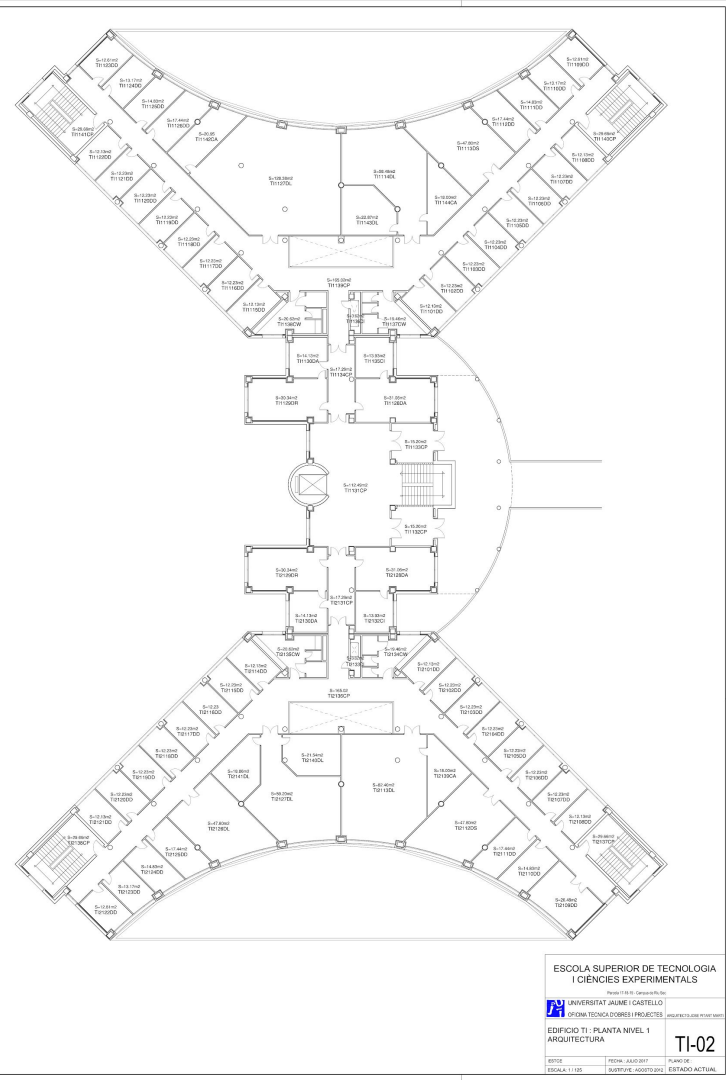
vertical: $0.841 * 125 / 6623$

```
origin: [ , , 0]
```

negate: 0

occupied_thresh: 0.65

```
free_thresh: 0.25
```



TI_n1.yaml

image: ./TI_n1.png

mode: trinary

resolution:

horizontal: $0.594 * 125 / 4678 = 0.0159$

vertical: $0.841 * 125 / 6623 = 0.0159$

origin: [, , 0]

negate: 0

occupied_thresh: 0.65

free_thresh: 0.25

Parameters

Map Server

Implements the server for publishing a map topic.

yaml_filename

Type	Default
string	N/A

Description

Path to map yaml file.

topic_name

Type	Default
string	"map"

Description

Topic to publish loaded map to.

frame_id

Type	Default
string	"map"

Description

Frame to publish loaded map in.

Publish and visualize the map of building T1

Download archive [Worlds.zip](#)

Uncompress it in your IR2121 repository folder.

Open a terminal window, cd to `Worlds/scripts` and run:

```
./visualize_map.bash
```

Open a terminal window, cd to `Worlds/scripts` and run:

```
./publish_map.bash
```

Interact

Move Camera

Select

Focus Camera

Measure

2D Pose Estimate

2D Goal Pose

Publish Point

+

-

Displays

Global Options

Fixed Frame

Background Color

map

48; 48; 48

Frame Rate

30

Global Status: W...

Fixed Frame

No tf data. Actual erro...

Grid

Map

Status: Ok

Topic

/map

Update Topic

/map_updates

Alpha

0,7

Color Scheme

map

Draw Behind

☐

Resolution

0,0159

Width

4678

Height

6623

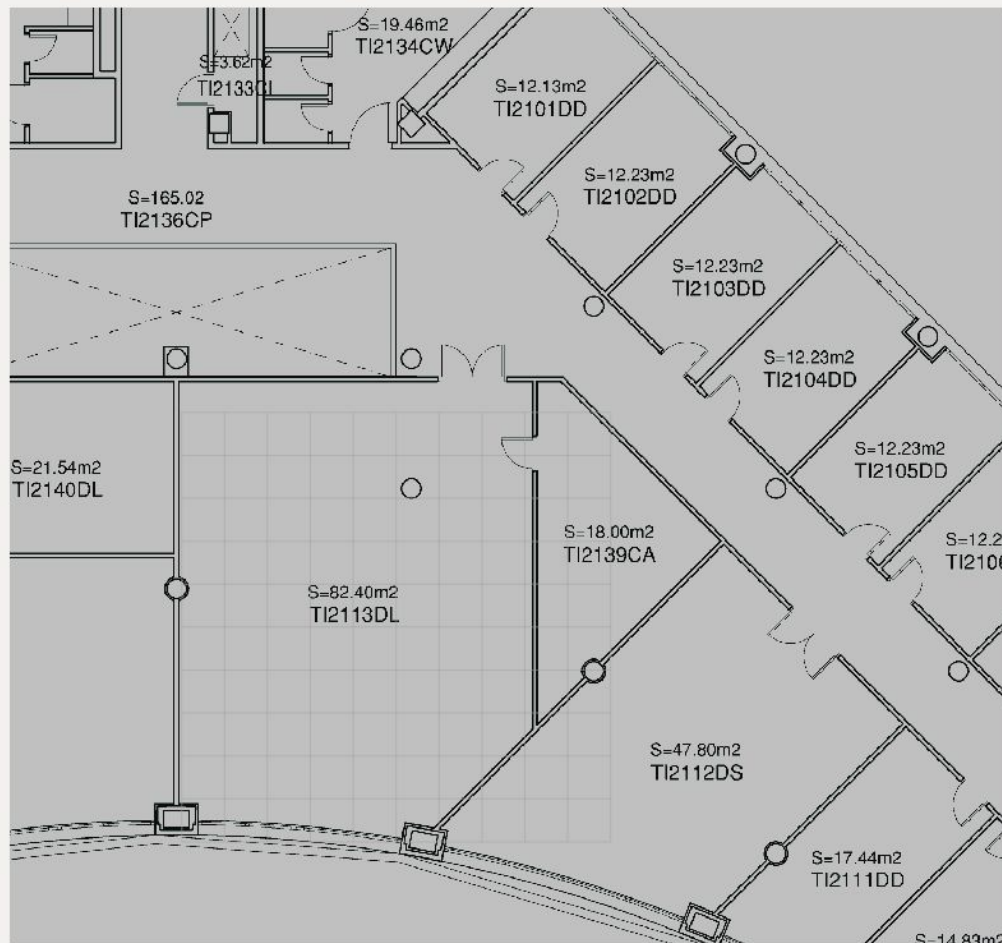
Position

-41.5; -28; 0

Orientation

0; 0; 0; 1

Use Timestamp

☐

Add

Duplicate

Remove

Rename

Reset

30 fps

Gazebo world

Exit the previous ROS scripts, and close those terminals.

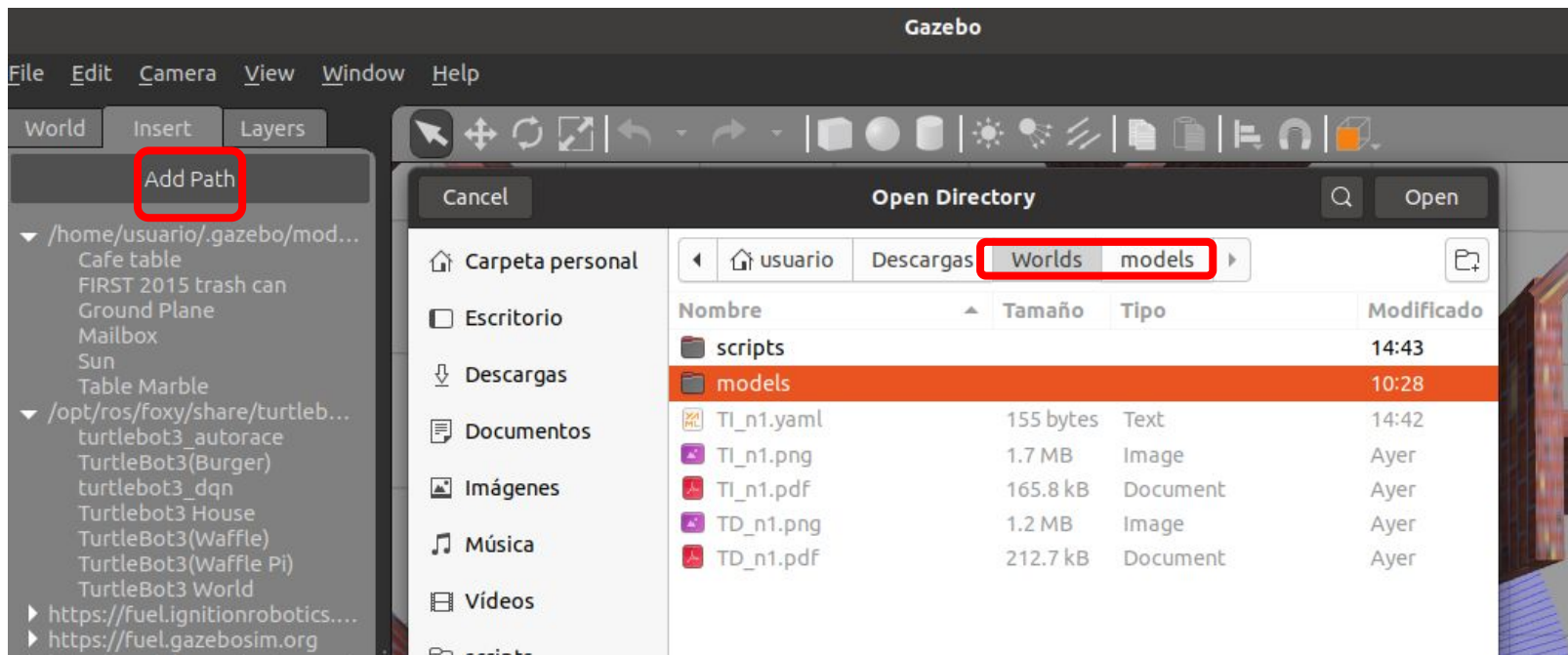
Open a new terminal window,

cd to `Worlds/scripts`

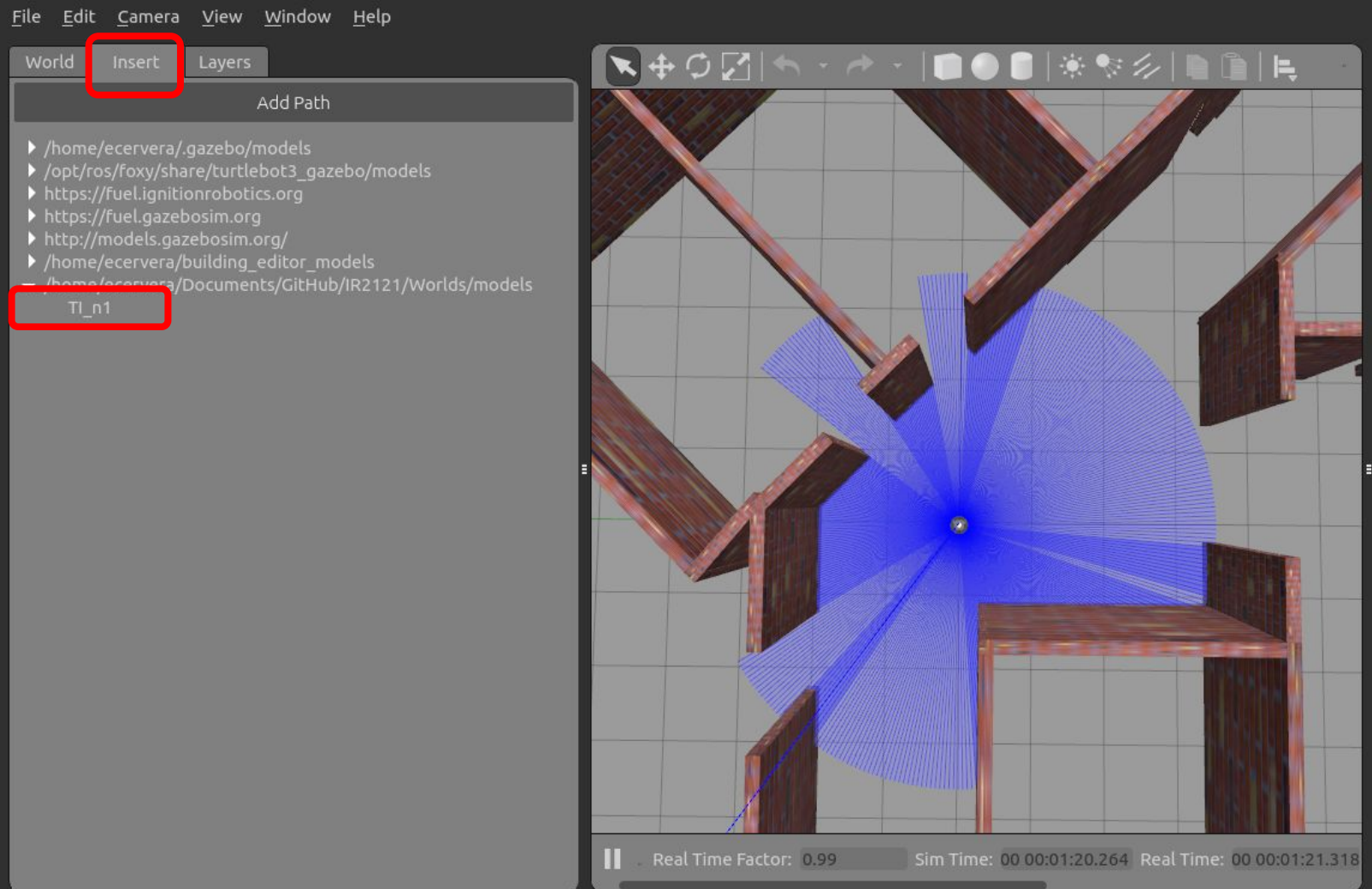
and run:

```
./launch_tb3_empty_sim.bash
```

Add
path
to
model



Insert model



Visualize robot and map

Open a terminal window, cd to IR2121/Worlds/scripts and run:

```
./visualize_robot.bash
```

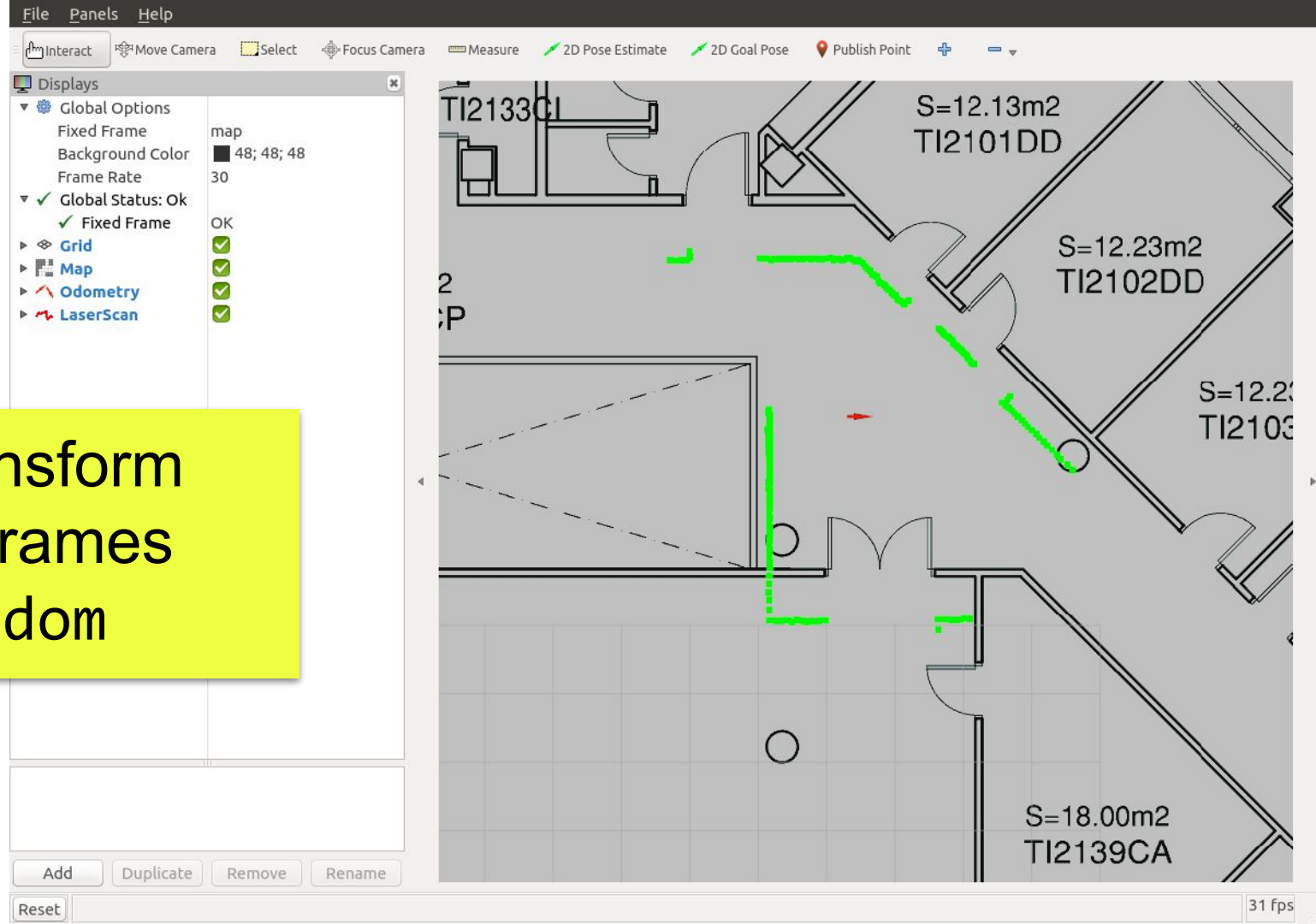
Open a terminal window, cd to IR2121/Worlds/scripts and run:

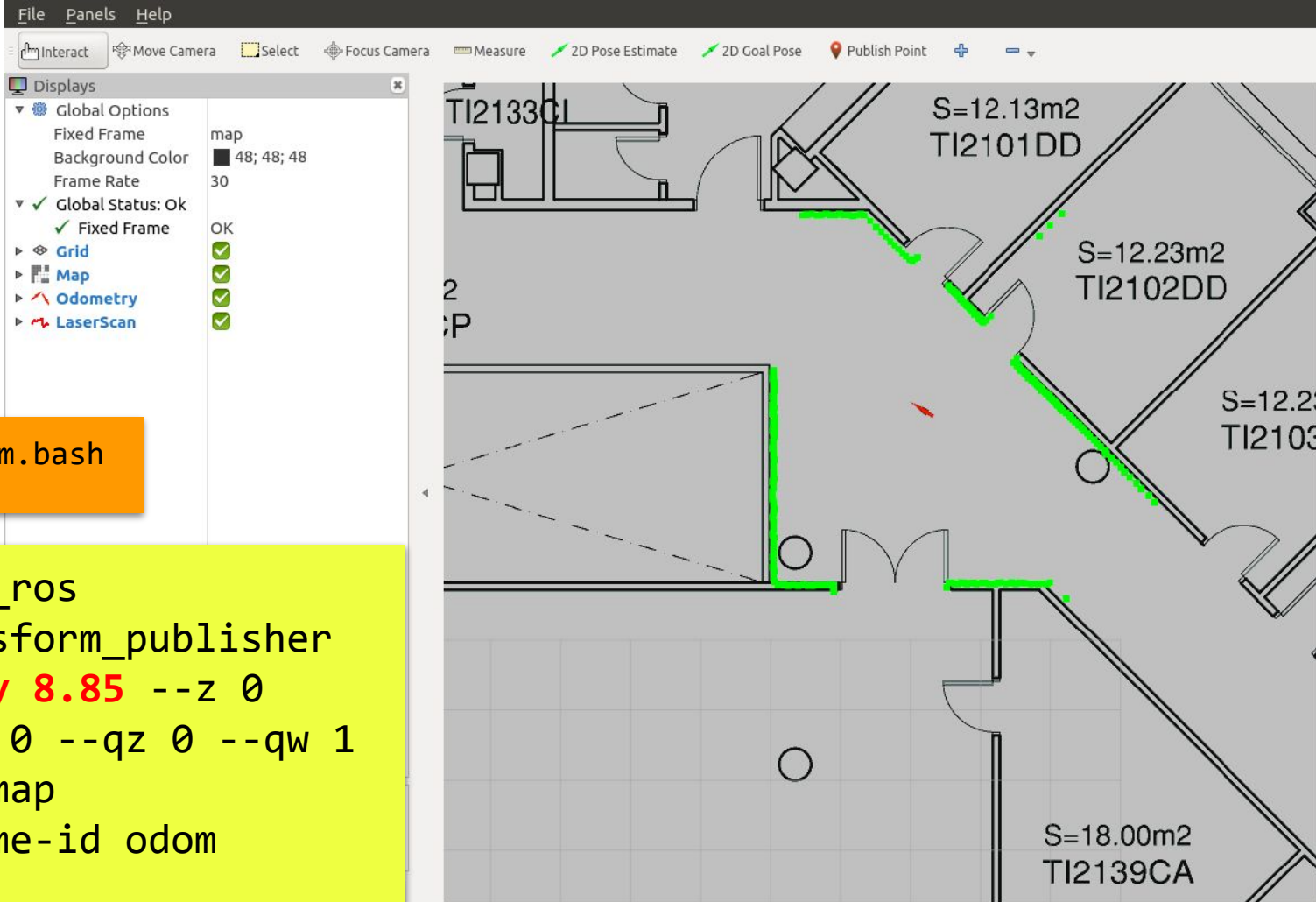
```
./publish_map.bash
```

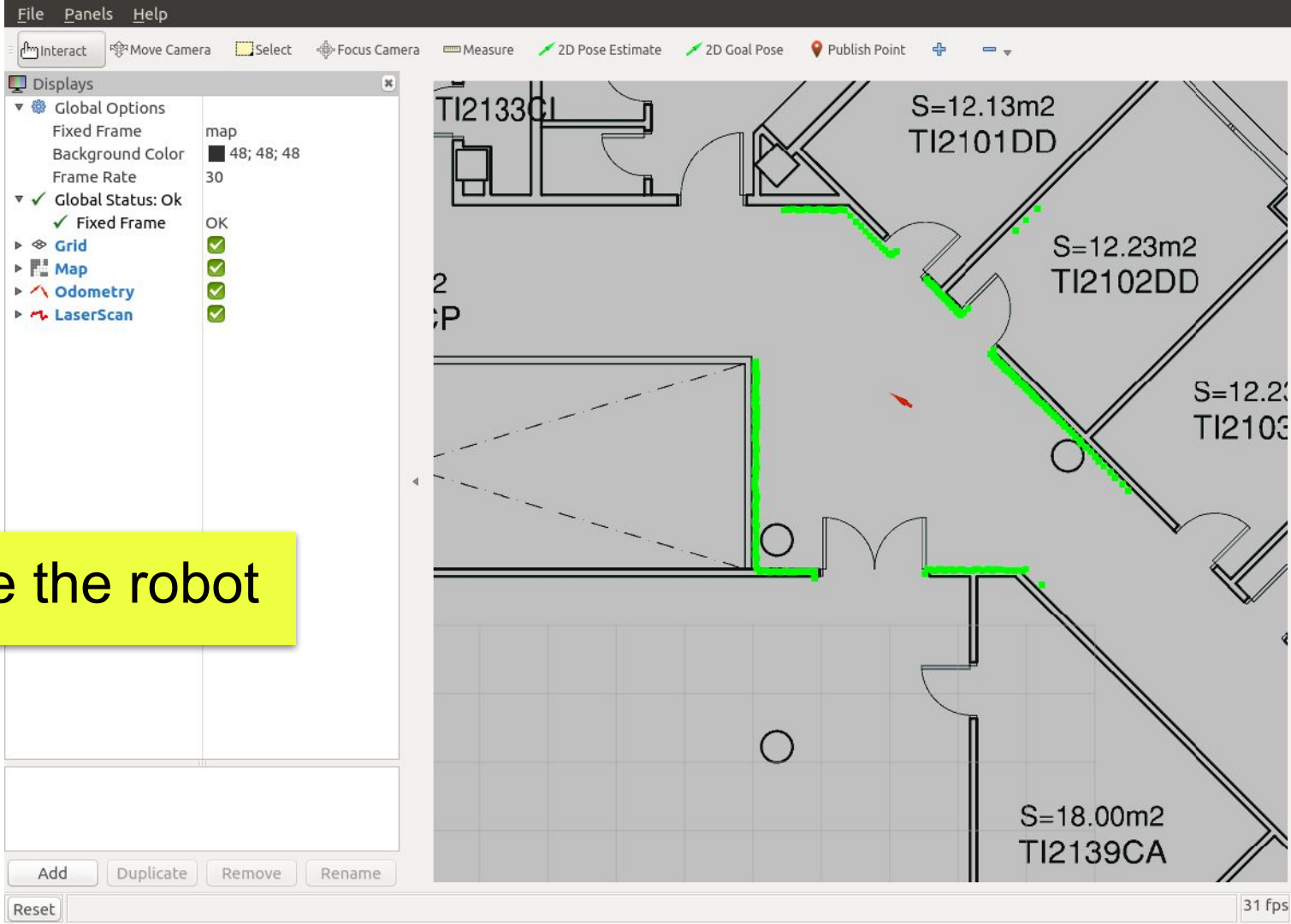
Open a terminal window, cd to IR2121/Worlds/scripts and run:

```
./publish_transform.bash
```

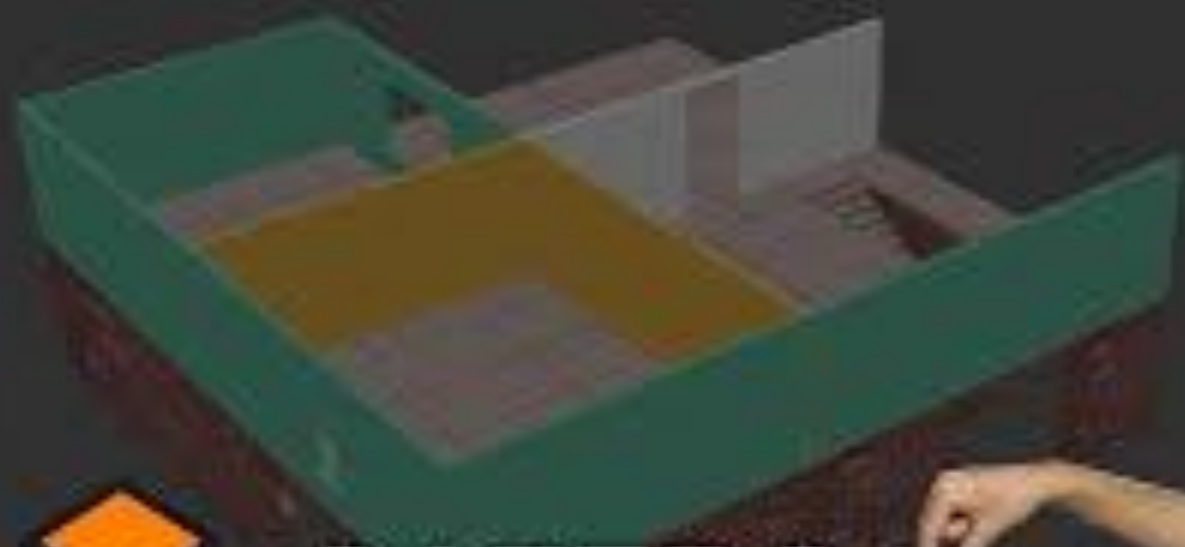
Adjust transform
between frames
map and odom







Teleoperate the robot

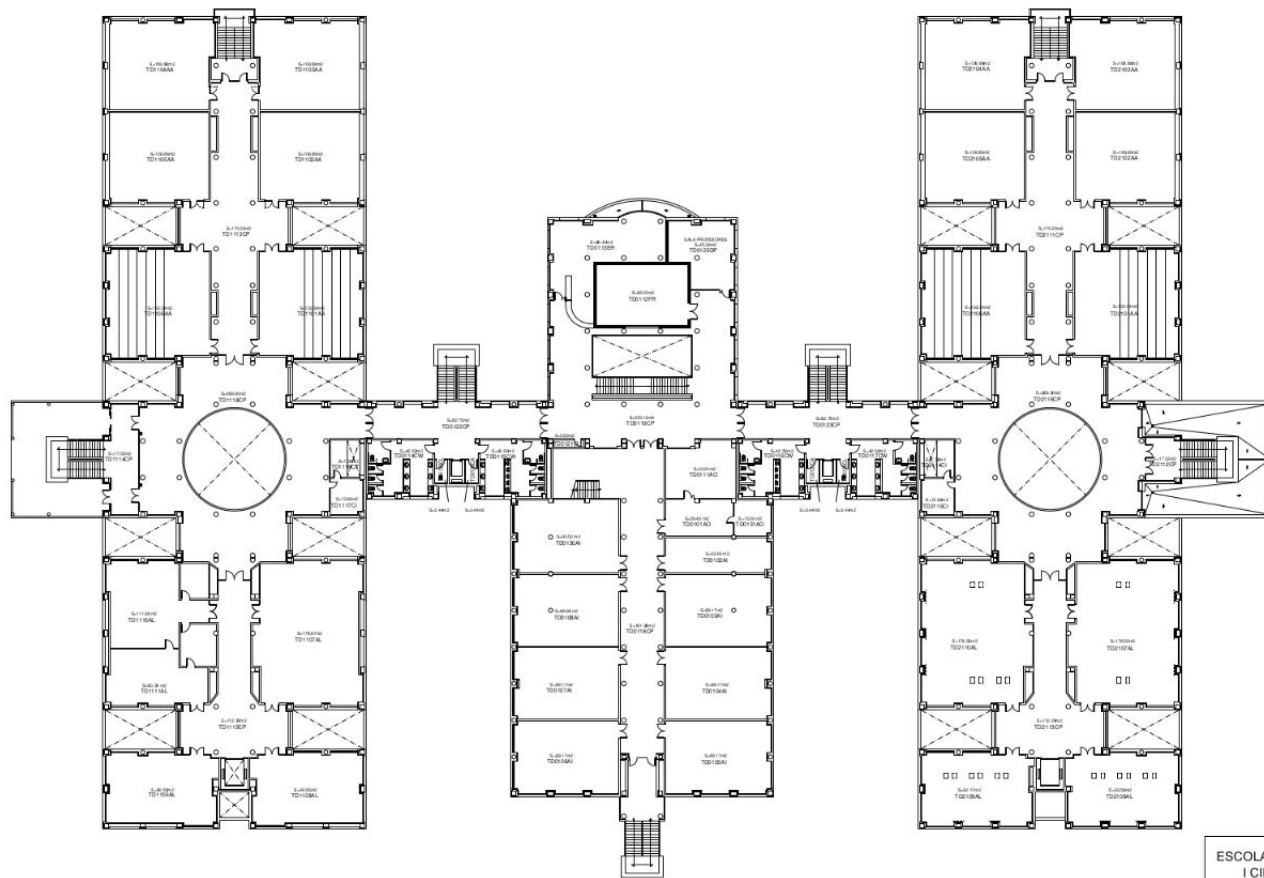


GAZEBO



BUILDING EDITOR





ESCOLA SUPERIOR DE TECNOLOGIA
I CIÈNCIES EXPERIMENTALS

Planta 11 de 19 - Campus de RiuSec



UNIVERSITAT JAUME I CASTELLÓ
OFICINA TÈCNICA D'OBRES I PROJECTES

ARQUITECTE: JORDI MARTÍ GARCÍA

EDIFICIO TD - PLANTA NIVEL 1
ARQUITECTURA

TD-02

EDICIÓ:
ESCALA: 1 / 200

FECHA: - SEPTIEMBRE 2022
SURTITULO: - JULIO 2018

PLANO DE:
ESTADO ACTUAL

TD_n1.pdf

Format: PDF-1.7

Paper size: A1, landscape (841 × 594 mm)

Scale: 1/200

[PDF to PNG](#)

Format: png (PNG)

Width: 6623 px

Height: 4678 px

TD_n1.yaml

```
image: ./TD_n1.png
mode: trinary
resolution: meters / pixel ???
origin: [ , , 0]
negate: 0
occupied_thresh: 0.65
free_thresh: 0.25
```

Task 1: simulation

Visualize the map of building TD, floor 1, in RViz.

Build a Gazebo model of the laboratory TD1107AL and the surrounding area.

Launch a Gazebo simulation of a TurtleBot 3 and the model of the laboratory.

Teleoperate the robot, and verify that the laser scan fits the walls in the map.

Move the robot and record the following topics in a bagfile for 60-90 seconds:

```
/clock /map /odom /scan /tf /tf_static
```

Submit the requested files in the corresponding task of Aula Virtual:

- `rosbag2_2025_...` (compressed bagfile)
- a screenshot of `rviz2` showing the map, the odometry and the laser signal

Task 2: physical robot

Visualize the map of building TD, floor 1, in RViz.

Launch the TurtleBot 3 physical robot.

Teleoperate the robot, and verify that the laser scan fits the walls in the map.

Submit the requested files in the corresponding task of Aula Virtual.

Move the robot and record the following topics in a bagfile for 60-90 seconds:

`/map /odom /scan /tf /tf_static`

Submit the requested files in the corresponding task of Aula Virtual:

- `rosbag2_2025_...` (compressed bagfile)
- a screenshot of rviz2 showing the map, the odometry and the laser signal

Tips for working with the physical robot

Use one of the laptops of the lab (WiFi or Ethernet)

Or, use the desktop PC with the Wifi PIROBOTNET6

Or, use your own laptop with a USB WiFi dongle / cable

Do **NOT** define `ROS_LOCALHOST_ONLY` (“unset”)

Do define `ROS_DOMAIN_ID`

Always check the values of the /odom topic

```
(ros2 topic echo /odom --no-arr)
```

Tips for working with the physical robot

Check the /scan topic in RViz

Reliability policy: 'Best effort'

Durability policy: 'Volatile'

Homework: model of building TD

[Build a more complete Gazebo model](#) of the TD central building, level 1.

For a satisfactory grade: include TD1107AL and TD1118CP.

For an excellent grade: also include TD2116CP.

Record a bagfile with the topics during a teleoperation of the robot across the building.

Submit these files:

- TD_n1.yaml
- TD_n1.zip (Gazebo model, including model.config and model.sdf)
- rosbag2_2025_xxx.zip (compressed bagfile)

References

[Using tf2 with ROS 2](#)

[map_server - ROS Wiki](#)

[Map Server / Saver — Nav2 1.0.0 documentation](#)

[Gazebo : Tutorial : Building Editor](#)